

Kempe Theory with Boundary: Relative Charge Quantization, Gluing, and the Frozen Disc

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Abstract

Kempe-change theory — which colorings of a triangulation reach which others by chain swaps — has been studied on closed surfaces since Fisk (1973); the modern literature (Mohar 2007; Mohar–Salas 2009; Feghali 2023; two papers posted November and December 2025) remains on closed graphs. We develop the relative theory: triangulated discs with pinned boundary colorings. Three results. (1) Relative charge quantization: with each 4-coloring of a triangulation one associates a ± 1 sign per face and an integer charge per vertex (the local degree of the coloring’s simplicial map to the tetrahedron boundary); at interior vertices the charge obeys the classical mod-3 closure and a quantization forced by degree parity; at boundary vertices the closure FAILS in a controlled way — the charge class mod 3 equals the cyclic displacement between the vertex’s two pinned boundary neighbors. A pinned boundary coloring is thereby a charge boundary condition in the literal sense. (2) Gluing consistency: when two discs are glued along a common boundary coloring, seam displacements cancel mod 3, charges add, and the closed-surface theory re-emerges; color-matched gluing is free for existence (Birkhoff) but Kempe dynamics composes only through relative reachability classes. (3) Relative reconstruction exactness: on a pinned disc the interior sign pattern determines the completion exactly — the $\mathbb{Z}_3$ and translation freedoms of the closed-surface reconstruction are anchored by the boundary, and simple connectivity kills monodromy. As a first exercise we exhibit a 19-vertex disc with Eulerian interior admitting two completions that are each relatively frozen (no legal move) and mutually unreachable — showing that Kempe-connectivity theorems for closed Eulerian triangulations do not transfer to Eulerian regions with boundary. (4) Boundary incompleteness (witnessed): the two frozen completions have identical charge vectors and sign patterns congruent modulo the full relative move span (computed exactly, dimension 17, no sampling) — no linear functional of the sign data distinguishes them,

so the sign-coset invariant, which is candidate-complete on the closed sphere, is PROVABLY incomplete on discs. (5) Non-monotonicity of freezing under deletion: puncturing can unfreeze — a 9-vertex closed example whose coloring space splits into two mutually unreachable classes becomes a SINGLE Kempe class after one vertex deletion. Together (1)–(5) organize into a topology trichotomy: closed sphere — linear-complete candidate; boundary — provably beyond linear algebra, witness in hand; higher genus — complete only with a cohomological monodromy summand. The boundary cell is where Kempe dynamics escapes linear algebra, and the 19-vertex disc is its Rosetta stone.

1. Setting and conventions

T a simple triangulation of the oriented sphere or of a closed disc; faces read counterclockwise. Colors = Klein group $V = \mathbb{Z}_2 \times \mathbb{Z}_2 = \{0, a, b, c\}$, reference cyclic order (a, b, c). Face sign $z_t \in \{\pm 1\}$: the parity of the edge-label triple (labels = endpoint color sums) read counterclockwise. Charge $c(v) = \sum_{t \ni v} z_t$; winding $\omega(v) = c(v)/3$ where defined. A Kempe swap on a chain S adds $g = p+q$ on S; its effect on signs is the Straddle-Flip law: z flips exactly on faces with 1 or 2 vertices in S. For a disc, the boundary cycle carries pinned colors; a move is legal iff its chain contains no pinned vertex.

2. Absolute background (closed sphere; proofs in the companion theorem note)

Theorem A (Charge Quantization). $c \equiv 0 \pmod{3}$ at every vertex (closed-walk winding of the link); $c \equiv \deg \pmod{2}$, $|c| \leq \deg$; hence $\omega = 0$ at deg-4, $\omega = \pm 1$ at deg-5 and deg-7, $\omega \in \{0, \pm 2\}$ at deg-6; $\sum_v c(v) = 12 \cdot \deg(f)$ with $\deg(f)$ the degree of the simplicial map $T \rightarrow \partial \Delta^3$. Odd-degree vertices admit no singleton chain (an odd link has no proper 2-coloring): knot sites are fixed by the graph; the coloring chooses only signs.

Theorem B (Reconstruction). On the sphere, z determines f up to the sign-preserving color group $A_4 \cong V \rtimes \mathbb{Z}_3$ (affine structure of V; odd color permutations flip all signs), and that group is Kempe-realizable; so equal sign patterns imply Kempe-equivalence. On surfaces of positive genus the reconstruction acquires an $H^1(\cdot; \mathbb{Z}_3)$ monodromy — the room where the Mohar–Salas mod-12 classes live; the invariant theory is genus-graded.

3. The relative theory

Theorem 1 (Relative quantization). Let D be a triangulated disc with pinned boundary coloring. At interior vertices Theorem A holds unchanged. At a boundary vertex v with fan link $u_{\text{first}}, \dots, u_{\text{last}}$ (endpoints pinned): $c(v) \equiv \text{disp}(f(u_{\text{first}}) \rightarrow f(u_{\text{last}})) \pmod{3}$, where $\text{disp} \in \{0, \pm 1\}$ is the cyclic displacement in the 3-cycle of colors $\neq f(v)$. Legal moves change $c(v)$ in even steps within its pinned class. *Proof.*

The winding argument on the open fan walk: the ± 1 face steps telescope to the displacement between the endpoint colors, which are pinned; interior fan vertices are free but alter the sum only by closed sub-loops $\equiv 0 \pmod 3$. Straddle-Flip changes each incident sign by ± 2 per flip. ■

Theorem 2 (Gluing consistency). Glue discs D_L, D_R along a common boundary coloring. At each seam vertex the two fans complete the closed link, and $\text{disp}_L(v) + \text{disp}_R(v) \equiv 0 \pmod 3$ automatically; hence $c(v) = c_L(v) + c_R(v) \equiv 0 \pmod 3$; the absolute closure re-emerges and charges add along the seam. Color-matched gluing is free for existence (any two properly colored copies of a triangle differ by a color permutation — the Birkhoff-era splitting); Kempe DYNAMICS composes only through the relative reachability classes of the pieces. *Proof.* The two fan walks traverse complementary arcs between the same pinned endpoints, so their displacements are inverse; existence gluing is classical; the dynamical statement is the definition of legality at the seam (a chain crossing the seam contains a pinned vertex). ■

Theorem 3 (Relative reconstruction exactness). On a pinned disc the interior sign pattern determines the completion exactly. *Proof.* A boundary edge's label is known and anchors the label propagation (killing the $\mathbb{Z}_3$ freedom); pinned colors kill the translation; the disc is simply connected, so propagation has no monodromy. ■

Consequently the relative Kempe-classification question is exactly the coset question for the sign pattern modulo the span of legal straddle indicators — with no symmetry bookkeeping at all. The Relative Availability-Saturation Conjecture (reachable $\mathbb{Z}$ equal coset) is the program's open frontier; its closed-sphere counterpart is consistent with an eight-graph exhaustive record at the time of writing.

4. First exercise: the frozen disc

There exists a 19-vertex triangulated disc with all interior vertices of even degree (Eulerian interior) and a pinned boundary coloring admitting two completions, each with NO legal Kempe move, mutually unreachable (exhaustive check). Hence: Fisk-type connectivity for closed Eulerian triangulations does NOT transfer to Eulerian regions with boundary; “Eulerian regions are free” is false relatively; and any decomposition-based approach to coloring theorems must glue DYNAMICS through relative classes (Theorem 2), not merely colors. By Theorem 3 the two completions have distinct interior sign patterns; whether they lie in distinct cosets of the legal-move span was decided by exact computation, with predictions lodged in a blind ledger beforehand.

Theorem 4 (Boundary incompleteness — witnessed). Define W_{rel} intrinsically as the $\text{GF}(2)$ span of straddle indicators of interior chains over ALL proper colorings of the disc (no reference to any pinning; a pinning-relative span degenerates to zero on frozen populations and forces vacuous verdicts). For the 19-vertex disc: the two frozen completions have identical charge vectors (interior and boundary), and their sign patterns differ by a weight-20 vector lying IN W_{rel} ($\dim W_{\text{rel}} = 17$, from 43 realizable interior chains, computed exhaustively). Hence no functional factoring

through the sign-coset — nor any linear functional of the charge data — separates two mutually unreachable completions: the sign-coset invariant is incomplete on discs. Combined with its closed-sphere candidacy, incompleteness is a property OF THE BOUNDARY, not of the invariant’s construction.

Theorem 5 (Freezing is not monotone under deletion). There is a 9-vertex sphere triangulation whose proper 4-colorings form two mutually unreachable Kempe classes, and a vertex whose deletion merges the coloring space into a SINGLE Kempe class (192 colorings; exhaustive). Puncturing can unfreeze. In particular the insertion problem of the classical induction — which lives in $G-v$ — inhabits a strictly friendlier regime than the closed graph itself.

5. Open problems

1. The separator problem (the Rosetta stone). Theorem 4 shows the residual relative invariant is not linear. What computable object separates the disc twins? Candidate classes under pre-registered exhaustive test: invariant quadratics of the sign pattern; counting (shadow) invariants mod 2; ordering-obstruction certificates (Hall-type blocked multisets in the word problem); boundary-refined static functionals; a groupoid-cocycle analogue of the genus cell’s monodromy summand. A total kill would itself be a theorem: the boundary obstruction would be genuinely dynamical, with no low-order computable shadow.
2. Characterize relatively-frozen completions (existence at 19 vertices with Eulerian interior is established; minimal examples? classification by boundary condition?) — and, per Theorem 5, characterize when deletion unfreezes.
3. The boundary-condition calculus: which charge boundary conditions (dis-vectors) admit completions at all; a relative existence theory beside the relative dynamics.
4. Genus-graded completeness: sign pattern $\oplus$ monodromy as the complete invariant off the sphere; the trichotomy’s third cell as a theorem rather than a mechanism.
5. Gate existence as a theorem (the closed-cell forcing lemma): every stuck insertion configuration admits a one-vertex-support 4-word — measured at scale, unproved; the first rung of the word-order descent program.

6. The height bridge (v0.3)

The relative theory embeds in the height-function formalism of lattice statistical mechanics (Peled–Spinka’s survey is the frame). Lift the Klein colors to $\mathbb{Z}^2$ by step vectors $A = (1,0)$, $B = (0,1)$, $C = (-1,-1)$ ($A+B+C = 0$; reduction mod 2 recovers the labels). Every properly colored face closes all-plus or all-minus, and coherence forces adjacent faces into opposite classes — a dual 2-coloring. Hence: a single-valued height exists iff the triangulation is Eulerian; at odd-degree vertices the height acquires monodromy — the charge theory’s knots are height DISLOCATIONS, and the winding $\omega(v)$ is the Burgers datum. On a pinned disc with Eulerian interior,

the pinning determines a boundary height walk whose total slope — the TILT — is a linear functional of the boundary word that no interior move can change: the tilt is the residual conserved quantity that Theorem 4’s twins made necessary, and “frozen $\mathbb{Z}$ extremal tilt” is the pre-registered mechanism under blind test at census scale. The three formalisms are one theory at three resolutions: heights ($\mathbb{Z}^2$) $\rightarrow$ charges (mod 3: the displacement classes of Theorem 1 are the height gradient’s mod-3 shadow) $\rightarrow$ colors (mod 2). The trichotomy follows in one stroke: closed sphere — heights exist iff Eulerian, else a dislocation gas (the charge dynamics); boundary — the tilt, carried by the pinning, invisible to interior invariants; higher genus — H^1 monodromy. The puncture asymmetry of Theorem 5 is explained: interior deletion cannot relax a boundary-carried tilt, and closed spheres have no tilt to carry.

7. Walls, confinement, and the hand-off (v0.4)

For two completions f, f' of one pinning, the difference field $\delta(v)$ — identity where they agree, the transposition $(f(v) f'(v))$ where they differ — decomposes the disc into DOMAINS of constant δ separated by WALLS. Two structural facts: (i) walls cannot terminate at interior vertices (link-cycle interface parity), so every wall is closed or ends on the boundary; (ii) dissolving a wall requires realizing a domain’s transposition by chains, and for boundary-anchored walls those chains cross the pinning — ANCHORING is the freeze mechanism. The 19-vertex twins realize the complete picture, measured: a flat, half-crystallized boundary (the filler, an iso-height wall — one color on a parity class $\mathbb{Z}$ constant boundary height); the filler color CONFINED to exactly one interior vertex (a monopole at the fixed center); two ground-state tilings exchanged by a wall-pair hinged on the monopole. The wall is irreducibly RELATIONAL: mod-2 sign functionals, invariant quadratics, and the full $\mathbb{Z}^2$ height-sector all take equal values on the twins (measured; the disc carries exactly three height sectors and the twins share one). Under the mirror hypothesis (a pinning symmetry exchanging the tilings — under census test) this blindness is forced: any functional built from the problem’s data commutes with its symmetries; the honest object is then a reachability METRIC, not an invariant.

Hand-off Theorem. In the insertion problem ($G-v$ with NO pinned vertices) no anchored wall exists between any two colorings at any step of any rescue walk: pinnedness is a datum of the problem, moves never create it, and anchoring requires it. The freeze mechanism witnessed by the twins is therefore structurally unavailable exactly where the classical induction lives. Scope, stated in the theorem’s own breath: this removes a mechanism; it does not grant a move — the existence of the rescuing gate (measured universally; dominant support three, the charge quantum in vertex form) remains the program’s open theorem, now shaped as defect dynamics: an elementary dislocation glide adjacent to an unpinned hole.

8. Scope, curl, alphabet, and the hand-off (v0.5)

Scope Theorem. The height lift of Section 6 exists exactly on knot-free (Eulerian) domains; on knotted domains the lift fails at every odd vertex, and the correct object is a flat connection with structure group $\mathbb{Z}^2 \rtimes \mathbb{Z}_2$ (translations and the checkerboard flip), with holonomy prescribed at the knots. Consequently: Kempe dynamics is height dynamics precisely where there are no knots; the residual — gate — dynamics lives at the knots. On knot-free simply connected domains the holonomy is trivial, which resolves the static separator program negatively and completely: all five candidate separator families (invariant quadratics, shadow counts, ordering-obstruction certificates, boundary-refined statics, and the holonomy cocycle — the last by derivation, since the frozen twins' domain is knot-free) fail, and the wall's irreducibly relational character stands as measured fact with a structural explanation-shape: nothing static was ever going to see it.

Gates. The measured unsticking moves displace ZERO charge: by the Straddle-Flip calculus their net re-signing is divergence-free at every vertex — gates are pure curls, and walls are what curls move. The measured patch alphabet has five letters (complement-of-one, triples, closed stars, 6-patches, charge-neutral singletons — the last derived: singletons exist only at neutral vertices); whether the alphabet collapses modulo global re-signing is under test. The open theorem of the program — Gate Existence — is now shaped as a Wall Motion Lemma: at every stuck insertion configuration some alphabet letter applies near the hole and strictly reduces wall-distance to a freed target.

Hand-off Theorem (final, even-degree form; banking pending one confirmation). The wall system of any pair of colorings is a GRAPH; at every interior vertex its degree is EVEN (the transition transpositions around a link cycle multiply to the identity; the S_4 sign homomorphism forces an even crossing count — this sharpened form of the interface lemma is due to the adversarial read, which improved the theorem it was auditing). Hence all odd-degree wall-graph vertices — in particular all endpoints — lie at topological boundary, and anchoring requires pinnedness, which no dynamical move (Constraint Persistence) and no proof operation (Surgery Persistence: deletion, WLOG edge-addition, restriction) can create. For the insertion problem ($G-v$, no pinned vertices), quantified over ALL pairs: no wall component is anchored. The freeze mechanism witnessed on the disc is structurally unavailable throughout the classical induction. Scope carved into the statement: the theorem removes a mechanism and does not grant a move (the Wall Motion / Triple Lemma is the successor obligation — the alphabet's one letter, a three-vertex pure-curl patch, conjectured to always apply near an unpinned hole with strict wall-distance descent); walls may branch and cross at even interior degree, and nothing downstream may assume simple curves; and the theorem does NOT protect pinned-seam proof architectures, where the freeze mechanism genuinely operates — the disc is the permanent witness for both halves of that sentence.

The Gauss law (exact, 240/240). For every completion of every filler pinning measured: $2 \cdot \text{Area}(\text{boundary height walk}) = -\sum_t z_t$ — the boundary winding reads

the enclosed total flux exactly. Flux-neutrality of the pinning is thereby a new NECESSARY condition for freezing (all 15 frozen pinnings are flux-neutral; so are 220 free fillers — sufficiency remains open: $\text{filler} \wedge \text{flux-neutral} \wedge \text{something finer}$).

Acknowledgments and provenance

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— internal draft; nothing here is banked until the standing review chain (Cal cold-read, toy verification, Casey’s word) completes.